

FULL-STACK AI & REAL-TIME DESKTOP FRAMEWORK

T-PYTREXT

PYTHON + RUST + ELIXIR + TAURI V2

34K+

LINES OF CODE

220+

CLASSES

9

DEMOS

96

TESTS (100%)

15

CLI COMMANDS



Architecture — 4 Powerful Layers

Layer	Technology	Purpose
1. Desktop UI	Tauri v2	Lightweight webview — 10x lighter than Electron, Rust security
2. Rust Core	PyO3 + Axum	Blockchain, AES-256, SQLx DB, Candle ML, Burn DL, MCP Server
3. Python Core	Python 3.12	220+ classes — LangChain, Hermes, RAG, HITL, Search, Events
4. Real-Time	Elixir BEAM	WebSocket, ClusterManager, TaskDispatcher — millions of connections



Performance Benchmarks

116,478/s Events per second	44ms App startup time
1,898/s Blockchain blocks/sec	10,892/s AES-256 encryptions/sec
359,274/s SHA-256 hashes/sec	7,543/s Async events (8 threads)

Key Features

Blockchain

SHA-256 immutable ledger. Create, verify, audit blocks. Tamper detection built-in. Perfect for finance, voting, supply chain.

AES-256 Encryption

Military-grade encryption. Encrypted SQLite database. Memory zeroization. Secure key management.

AI/ML Engine

LangChain + Hermes Agent + PyTorch + Candle + Burn + RAG. Build intelligent apps with one line.

Real-Time

Elixir BEAM VM handles millions of concurrent connections. Perfect for chat, trading, live dashboards.

MCP Protocol

Model Context Protocol — Server + Client. Expose tools to AI models. Connect to external MCP servers.

Human-in-the-Loop

Approval workflows for AI actions. Timeout management. Audit trail. Tauri notifications.

Web Search

SearXNG + DuckDuckGo integration. Multi-engine search with automatic fallback. RAG-ready.

Desktop UI

Tauri v2 lightweight webview. No Chromium bloat. Cross-platform: Windows, macOS, Linux.

9 Live Demo Projects

#	Demo	Category	Key Features
1	 Smart Bank POS	FinTech	Blockchain, AES-256, AI Signatures, SQLx
2	 Smart Retail POS	E-Commerce	YOLO AI, Real-time Inventory, Elixir
3	 Chat System	Social	WebSocket, E2E Encryption, Groups
4	 Crypto Trading Bot	Trading	AI Market Analysis, Blockchain, HITL
5	 Hospital Management	Health	Encrypted EHR, AI Diagnosis, Prescriptions
6	 AI Exam System	Education	RAG, Hermes AI, Auto-Grading
7	 Real Estate Platform	Property	AI Valuation, Contracts, M-Pesa
8	 Blockchain Voting	Government	Immutable Votes, Anti-Fraud, HITL
9	 CyberSec Toolkit	Security	Port Scanner, OSINT, Forensics, AI Audit

Deployment — Any Server

 Docker One command: <code>docker-compose up -d</code>	 Kubernetes <code>kubectl apply -f k8s-*.yaml</code>
 AWS Lambda Serverless — pay per request	 GCP Cloud Run 2M requests/month FREE
 Linux VPS <code>bash deploy-server.sh</code>	 Windows Server PowerShell <code>install-service.ps1</code>

15 CLI Commands

Command	Description
pytrex init <name>	Create new project
pytrex dev	Dev mode with hot-reload
pytrex scan	Discover all PyTreXT projects
pytrex run <name>	Run any project
pytrex test	Test all modules (96 tests)
pytrex deploy	Deploy to production
pytrex dashboard	Open project dashboard
pytrex build	Build for production

License & Pricing

Tier	Price	Who
 Community	FREE	Students, Startups <\$1M, Non-profits, Personal
 Startup	\$499/yr	Small companies, SaaS, Agencies (≤10 devs)
 Professional	\$2,999/yr	Banks, Government, Healthcare, Mid-size (≤50 devs)
 Enterprise	Custom	Large corporations, Military, Telecom, Global
tz Tanzania	TZS 250K-1.5M	Special local pricing for Tanzanian companies

Framework at a Glance

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Languages (Python+Rust+Elixir)

15

Deploy Configs (Docker,k8s,Cloud)

